HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented REST APIs using **Java Spring Boot** for seamless data exchange between microservices, handling approximately 500 requests per second during peak loads.
- Developed and maintained Python scripts for automating data ingestion and transformation tasks, reducing manual processing time by **40%** for routine reporting.
- Engineered efficient SQL queries and database schemas across **PostgreSQL** and **Oracle** for data validation and ad-hoc reporting, supporting critical financial operations.
- Built and supported production-quality AI-powered workflow automations using **Python** and **Rust**, integrating with enterprise applications to reduce manual activity by an estimated **15%**.
- Developed production-quality solutions using **Python** and **Rust**, with exposure to **TypeScript** and **Node.js** where appropriate, to improve business processes.
- Supported the development of AI agent workflows, including prompt design, tool use, orchestration, and evaluation, implementing appropriate technical safeguards.
- Containerized applications using **Docker** and deployed them to **AWS EC2** instances, managing CI/CD pipelines through **Jenkins** for streamlined deployments.

SKILLS

AI/ML – Generative AI, LLM, Prompt Design, AI Agents, Tool Use, Orchestration, Function Calling, RAG, Semantic Search, Vector Databases

Languages – Python, Rust, Java, TypeScript, Node.js, SQL

Frameworks/Libraries – Spring Boot, FastAPI, scikit-learn, pandas, NumPy

Cloud/DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git, GitHub, CI/CD

Databases – PostgreSQL, Oracle, MySQL, Dataverse

Microsoft Technologies – Microsoft 365, SharePoint, Power Platform, Microsoft Graph

Methodologies – Agile, Scrum, REST API Design, Software Engineering Practices

PROJECTS

AI-Powered Workflow Automation Platform

- Developed a core AI agent framework in **Python** and **Rust** to orchestrate multi-step workflows, integrating with enterprise systems via **REST APIs**.
- Implemented prompt engineering techniques and tool-use capabilities for the AI agents, enabling them to interact securely with **Microsoft Graph** and **SharePoint** data.
- Containerized the AI services using **Docker** and deployed them to an **AWS** environment, ensuring scalability and reliability for internal business process improvements.

Customer Integration API for Enterprise Systems

- Built a secure **REST API** using **Java Spring Boot** to facilitate data integration between a client's legacy systems and a new enterprise platform.
- Designed and implemented robust error handling and authentication mechanisms, processing approximately **3,000** transactions per day for a key enterprise customer.
- Authored comprehensive API documentation and provided post-implementation support to the client's technical team, ensuring smooth adoption and integration.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)